# School of Advanced Computing & Information Technology

Department of Computer Science, Gujarat University

## MCA123 Data Analytics

## Assignment - Unit 5 & 6 (Practical)

**Unit 5 and 6** (Book name- “Statistics for Business & Economics” by Anderson)

| Title | Content | Book Chapter | Example No. |
|---|---|---|---|
| Probability | basic relationships of probability | 4 | 24,25,27,30,31,32 |
| Probability distribution | Binomial, Poisson, Hyper geometric, Normal and standard normal | 5 and 6 | Chapter 5 – 28,29,31,32,33,40,41,42,49,50,51<br>Chapter 6- 17,18,19 |
| Hypothesis Testing | Sigma known and unknown case | 9 | 16,17,18,20,28,30,33,34 |

1. Plot the probability mass function (PMF) for the number of heads from 0 to 10.
2. Calculate and plot the cumulative probability $P(X \leq x)$ for all x from 0 to 10.
3. The average number of cars passing through a toll booth per minute is 3. Find the probability that exactly 5 cars pass through in a minute using the Poisson distribution.
4. Plot the Poisson PMF for $x = 0$ to 10 for $\lambda = 3$.
5. Repeat Q5 for $\lambda = 5$ and $\lambda = 7$. Compare the shape of the distributions.
6. Plot the normal distribution curve and shade the region between 60 and 80.
7. Compute the probability that a value from a standard normal distribution is less than 1.96.
8. Find the z-value corresponding to the top 5% of a standard normal distribution (i.e., the 95th percentile).
9. Plot the standard normal curve and highlight the critical regions for a 95% confidence level.